Human impact

ACTIVITIES BY HUMANS CAN CHANGE ECOSYSTEMS AND THE PLANTS AND ANIMALS WITHIN THEM.

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Scientists know there have been five mass extinctions in Earth's history, all with natural causes. Many of today's species are becoming extinct due to human activities. Some experts think we are living through a sixth mass extinction right now.

Habitat loss

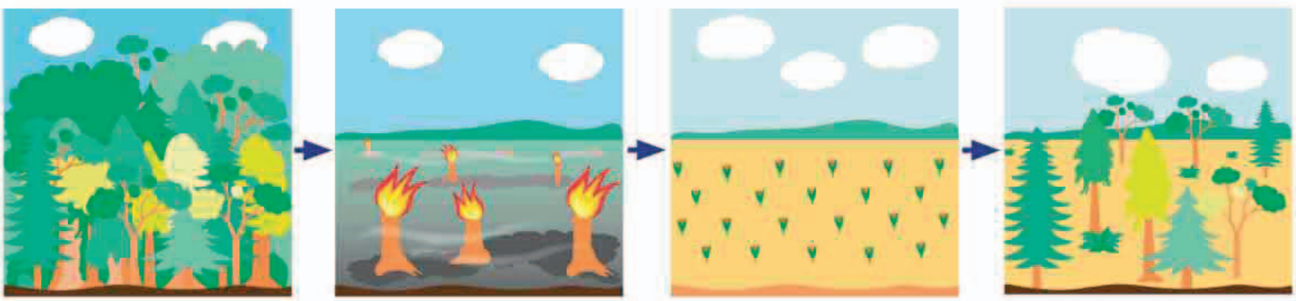
Humans have the ability to alter a habitat to suit their needs, turning natural landscapes into artificial ones, such as farmland or urban developments. The wildlife from the original habitat has been evolving for millions of years, and its species are specialized to a life within that community. They cannot survive in other habitats, and sometimes face extinction as a result.

▽ **Climax habitat**
Climax habitats carry the maximum number of life forms possible. This patch of tropical forest has a unique community of species.

▽ **Slash and burn**
Humans need places to grow crops, so they cut down the forest and burn the logs. The ash makes a nutrient-rich soil for the first crops.

▽ **Fertile soil**
For a few years the ashy soil makes good farmland. However, the orginal jungle soil beneath does not hold nutrients for long, so eventually the crop yields fall.

▽ **Secondary forest**
The farmers abandon this plot, and move on, leaving a new forest habitat to develop. It will never recover to its original climax state.



Fragmentation

Many forest animals never leave their habitat. For example, the gibbons of Southeast Asia can walk only short distances on the ground—they move about by swinging from branch to branch. Even a narrow gap in the forest, such as a road, is enough to divide forest communities permanently. Fragmentation breaks forest animals into small groups, making it harder for them to survive.

▷ **Living with relatives**
The biggest problems caused by fragmentation are loss of diversity and inbreeding. In a small group, every member is closely related to each other. Relatives share the same genes and any offspring tend to be weak.